

Arcesium(Internship):

Online Test:

- 1) Aptitude test- Questions were lengthy but involved basic concepts like Profit,Percentage and Ratio. Need to practice to solve in time.
- 2) Technical MCQs- Questions related to Debugging, Find the Output for programs in C++,Java were asked. OOPS, OS and DBMS were also asked.
- 3)2 coding questions- one was BFS traversal to find maximum in a level and other was a math based question.

Interview

Round-1

- 1) One coding question:

Given an array of elements with duplicates, print all the distinct permutations of the array. The interviewer expected the logic and wanted to know how I handle the duplicates in the permutations produced.

I was asked to code and explain the flow with an example.

Round-2

Technical discussion:

- 1)Qns related to OS:

- Windows vs Linux
- Steps from C program to a runnable executable
- Multithreading
- Memory,how processes are scheduled and about ready queue
- Can a programmer control threading options and make a program to run in specified number of threads? What is the maximum number of threads that you can spawn?
- Diff bw process and threads
- Role of OS, how it provides security

- 2)DBMS

- Transactions
- Why Normalization
- Why we need DBMS,can't OS do itself?
- How durability is ensured ?

Round-3

Technical discussion:

- 1)Qns related to OS again:(Your interested subject will be asked and most probably questions will be asked from that)

- Deadlocks
- Page fault
- Demand paging
- File structure/INODEs

2)OOPS basic qns

- Virtual functions and what is its use
- Virtual constructor and destructor
- Use of Protected access modifier
- Friend functions

3) Given a file, the size of it is unknown, how will you access the last 50 lines of it?

Optimized single traversal approach was expected.(Check 2 pointer approach of singly linked list)

4)DBMS

- 2 different database tables, how a single transaction works across tables?
- What commands to use to work with Transactions? TCL -commit,rollback,savepoint

There was no HR round.

Tips:

- Focus on the approach and the way you start with a problem.
- Keep interacting with interviewer and keep telling your thoughts,they need to understand how you think and how the solution evolves in your mind.
- Use whiteboard or the editor itself to explain,if you want to explain your logic.
- Be confident on your approach and its fine even if you aren't sure of the answers,just tell them that you aren't very sure of the concept.

D.E Shaw

Online Test:

1)Aptitude section - simple aptitude problems, Indiabix problems should suffice. Age,blood relation,directions

2)Tech MCQs- mostly Code outputs and Debugging. They didn't stick with any programming language and gave just the pseudocode.

Questions from OS/DBMS and DSA were asked.

3)2 coding questions

Interviews:

Round 1:

- 1) Given a function that'll output 1 and 0 with equal probability(50-50), design a function that will output 1 ,75% of the times and 0 ,25 % times.
I suggested OR ing the outputs of the function.
- 2) Given a binary string (01001), find contiguous subarray with equal no.of 0s and 1s.The 1s and 0s should be contiguous as well.
Ex:01,0011 etc

3) Given 2 tables as 2D arrays, with first row as column headers, design an algorithm to compare the given tables are identical.

Variations:

If the rows are shuffled. If both rows and cols are shuffled.

4) Questions about Scheduling and various types of Normalization types.

5) Given a binary tree, give its pre, post and inorder traversal.

Round 2:

Tech discussion:

- 1) A 3x3 array is given and a rule to traverse the array left to right was specified. We can start from any element in the first column and traverse till the rightmost column summing the elements along the path. We need to find the path with maximum sum. Time was given to try out possibilities and they allowed me to experiment before starting to code.
- 2) An array with only one duplicate number is given and I need to find the number. Different approaches were discussed and they asked the Complexity of each.

Questions related to OS and DBMS were also asked.

OOPS- Virtual functions, Abstract classes, and their necessity.

I had No HR round.

Suggestions:

- 1) DE Shaw followed a rigorous process and each round had 2 Interviewers and they were ready to spend time and listen to all our ideas and thoughts before coding.
- 2) They wanted you to be sure of the role you are applying for. They asked at the start of each round about the role and made sure I was clear about it.
- 3) Rounds were conducted in Hackerank pair coding site and it is good to use the whiteboard or the editor to explain your ideas.
- 4) Be strong on your Problem-solving skills and that can help when you encounter a new problem.

All the very best !!

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